
When Does Musical Structure Help?

A GNN–BERT Study of Context Understanding

Across Five Music Datasets

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Course: Neural Networks (CSE425 / EEE474 / CSE715)

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Abstract

Music carries context in two loosely coupled channels: a *relational* channel (chord transitions, repeated segments, temporal structure) and a *semantic* channel (captions, tags, artist and album metadata). We build a hybrid system that encodes the first with a graph neural network over music-structure graphs and the second with BERT, and we fuse them for multi-label context prediction. Rather than reporting a single headline number, we run the same four models across five datasets—FMA-small, MagnaTagATune, GTZAN, DEAM and MusicCaps—to ask a sharper question: *when* does structure help? The answer is not uniform, and that is our main finding. Graph features beat a mel-spectrogram CNN by 19 macro-F1 points on GTZAN but lose to it on FMA-small. Cross-attention fusion is the best of four fusion variants on DEAM (0.429 macro-F1, versus 0.282 for BERT-only and 0.181 for GNN-only) yet is *beaten by a GNN with no text at all* on MagnaTagATune (0.352 versus 0.390). The auxiliary emotion head only works when the graph is present: valence R^2 is 0.52 with cross-attention and -0.13 without the graph. A contrastive dual encoder reaches $R@5 = 0.193$ on MusicCaps caption→audio retrieval, $20\times$ the random reference, but its zero-shot tagging remains far below the supervised model (0.124 versus 0.429). We conclude that the value of music-structure graphs is governed by whether the target labels describe structure, and we report the negative results that support this.

1 Introduction

A single music clip expresses several kinds of context at once: genre and era, mood and emotion, harmonic structure, lyrical semantics and timbral texture. Sequence models over spectrograms capture local texture well but are poorly suited to *relational* structure—how a chord progression $C \rightarrow G \rightarrow Am$ recurs, or how two distant segments of a track are near-duplicates. Graph neural networks are a natural fit for that relational channel, and BERT is a natural fit for the textual channel of tags, captions and metadata.

This report implements and evaluates the four-task roadmap of the project brief: (1) a BERT multi-label tag classifier over text alone; (2) a GraphSAGE/GAT encoder over music-structure graphs built from audio alone; (3) a cross-attention GNN–BERT fusion with an auxiliary valence/arousal regression head; and (4) a contrastive dual encoder aligning audio graphs with natural-language captions.

Our contributions are:

Table 1: Datasets, after preprocessing. “Graphs” counts tracks for which a music-structure graph was successfully built; failures are corrupt source files (e.g. the well-known truncated `jazz.00054.wav` in GTZAN).

Dataset	Clips	Graphs	Tags	Text channel
FMA-small	8,000	7,997	50 (8 top genres)	title, artist, album, tags
MagnaTagATune	21,361	21,358	50	artist/album/track from path
GTZAN	1,000	999	10	<i>generated</i> from audio descriptors
DEAM	1,795	1,795	20 + valence/arousal	title, artist, album, last.fm
MusicCaps	5,521	5,161	50 aspects	expert free-text captions

1. A single reproducible pipeline that builds chord- and segment-graphs for five datasets (56,589 graphs total) and trains all four tasks with identical hyper-parameters and split discipline.
2. A **cross-dataset comparison** of each task, which is what exposes the main result: the benefit of graph structure is dataset-dependent and predictable from the semantics of the label space.
3. Honest reporting of three negative or partially negative results—cross-attention is not uniformly the best fusion; the graph-coherence diagnostic saturates and is uninformative as specified; and zero-shot transfer from the contrastive space is far weaker than supervision.

2 Problem Formulation

A track is a tuple $\mathcal{T} = (\mathbf{X}_{\text{audio}}, \mathbf{X}_{\text{text}}, G, \mathbf{y})$, where $\mathbf{X}_{\text{audio}}$ is a log-mel/chroma representation over T frames, $\mathbf{X}_{\text{text}}$ is a tokenised caption or metadata string, $G = (\mathcal{V}, \mathcal{E})$ is a music structure graph, and $\mathbf{y} \in \{0, 1\}^K$ is a multi-label context vector.

The text encoder maps tokens to contextual states $\mathbf{H}_{\text{text}} = \text{BERT}(\mathbf{X}_{\text{text}}) \in \mathbb{R}^{L \times d}$ with CLS vector $\mathbf{t} = \mathbf{H}_{\text{text}}[0]$. The graph encoder applies L rounds of message passing,

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\mathbf{W}^{(l)} \cdot \text{CONCAT}\left(\mathbf{h}_i^{(l)}, \text{MEAN}_{j \in \mathcal{N}(i)} \mathbf{h}_j^{(l)}\right)\right), \quad (1)$$

followed by a permutation-invariant readout $\mathbf{g}$. A fusion function combines the two, and a sigmoid head predicts tags:

$$\mathbf{z} = \text{Fusion}(\mathbf{g}, \mathbf{t}), \quad \hat{\mathbf{y}} = \sigma(\mathbf{W}\mathbf{z} + \mathbf{b}). \quad (2)$$

The objective is binary cross-entropy per tag plus an optional auxiliary term,

$$\mathcal{L} = - \sum_{k=1}^K \left[y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k) \right] + \alpha \|\mathbf{v} - \hat{\mathbf{v}}\|_2^2 + \beta \|\mathbf{a} - \hat{\mathbf{a}}\|_2^2, \quad (3)$$

with $(\mathbf{v}, \mathbf{a})$ the DEAM valence/arousal targets where available.

3 Data and Preprocessing

3.1 Datasets

Table 1 summarises the five datasets and how each is used. We satisfy the brief’s requirement of at least one primary audio dataset and at least one text/tag source several times over, and we use the advanced pairing (FMA/MagnaTagATune combined with DEAM and MusicCaps).

Two dataset-specific decisions materially affect interpretation and are stated up front rather than buried:

GTZAN has no metadata. The only text attached to a GTZAN clip is its filename, which *is* the label (`blues.00000.wav`). Using it would leak the target and produce a meaningless near-perfect score. We instead synthesise a descriptor from the hand-crafted features shipped with the dataset—tempo, spectral centroid, RMS, zero-crossing rate, spectral bandwidth and chroma spread—binned into natural-language phrases, e.g. “*A very fast instrumental excerpt at about 152 beats per minute. The*

timbre is bright and spread across the spectrum...” This text describes the audio, not the label, so the task is legitimate; but it is machine-generated, so GTZAN’s Task 1 score is **not comparable** to MusicCaps’ human captions.

MusicCaps aspects are quoted in the captions. The top-50 aspect phrases used as Task 1 labels appear near-verbatim in the caption text. The brief itself calls this a “caption $\rightarrow$ tag proxy task”. Its high score therefore measures extraction, not inference, and we mark it as such throughout.

3.2 Audio front-end and graph construction

Audio is resampled to 22,050 Hz mono and peak-normalised per track. We compute a log-mel spectrogram (128 bins), MFCCs (20), chroma-CQT (12), spectral contrast (7) and five scalar descriptors (centroid, bandwidth, rolloff, ZCR, RMS). Tracks are cut into 3 s windows with 1.5 s hop over the first 30 s, capped at 24 segments; MusicCaps clips are only 10 s, so they use a 2 s/1 s window (12-segment cap). A beat-synchronous alternative using `librosa`’s beat tracker is implemented and selectable.

Each segment is a node with an 89-dimensional feature vector

$$\left[\underbrace{\text{mfcc}_{\mu}(20)} \underbrace{\text{mfcc}_{\sigma}(20)} \underbrace{\text{chroma}(12)} \underbrace{\text{contrast}(7)} \underbrace{5 \text{ scalars}} \underbrace{\text{chord one-hot}(25)} \right].$$

Chords are estimated by cosine matching each segment’s mean chroma against 24 major/minor triad templates plus a “no chord” symbol gated on low energy.

Three edge families are merged into one graph, with edge attributes `[is_temporal, cosine_similarity, chord_transition]`: (i) temporal adjacency $i \rightarrow i+1 \dots i+r$ with $r = 2$ and weight $1/d$; (ii) similarity edges to the top $k = 5$ neighbours with cosine similarity $> \tau = 0.7$; and (iii) chord-transition edges between segments whose estimated chords form an observed transition, weighted by transition count. The resulting graphs average 19.0 nodes and ≈ 133 edges (mean degree ≈ 7) for 30 s tracks, and 10.0 nodes / 49.8 edges for the 10 s MusicCaps clips.

3.3 Splits and leakage control

FMA and MagnaTagATune ship official splits and we use them verbatim. MusicCaps and DEAM receive a seeded 80/10/10 split **grouped by artist**, so no artist appears in two splits; GTZAN, which has no artist metadata, is split stratified by genre. A leakage audit is written to `data/splits/leakage_report.json`.

One caveat is worth stating because it bounds the MagnaTagATune numbers: the *standard* MagnaTagATune directory split (folders 0–b / c / d–f) is **not** fully artist-disjoint—57 of 229 artists straddle splits. We keep it because the brief asks for the official split and because it is what published numbers use, but an artist-safe alternative is available via a flag. GTZAN is known to contain repeated artists and duplicated excerpts [Sturm, 2013]; with only 100 test clips its numbers carry real variance, and we treat differences below ~ 5 points on GTZAN as noise.

4 Models

Task 1 (BERT tagger). DistilBERT [Sanh et al., 2019] with a linear head on the CLS vector, $\hat{y}_k = \sigma(\mathbf{w}_k^\top \mathbf{t} + b_k)$, trained with BCE per tag.

Task 2 (GNN). Three convolutions are compared—GraphSAGE [Hamilton et al., 2017], GAT [Veličković et al., 2018] and GCN [Kipf and Welling, 2017]—each with 3 layers, hidden width 128, batch normalisation and dropout 0.3. The readout concatenates mean and max pooling. Node features are standardised using **training-split statistics only**.

Task 3 (fusion). Four variants are trained under identical conditions: `bert_only`, `gnn_only`, `concat` (early concatenation of `g` and `t`), and `cross_attn`, in which the graph readout queries the BERT token states,

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right), \quad \mathbf{Q} = \mathbf{g}\mathbf{W}_Q, \mathbf{K} = \mathbf{H}_{\text{text}}\mathbf{W}_K, \quad \mathbf{z} = \text{CONCAT}(\mathbf{g}, \mathbf{A}\mathbf{H}_{\text{text}}). \quad (4)$$

Table 2: Task 1, DistilBERT on text only. All four datasets beat the random-tag baseline by a wide margin. The train column exposes the failure mode: FMA-small memorises artist and album strings and generalises poorly across an artist-disjoint split.

Dataset	n_{train}	K	Train F1	Val F1	Test F1	Test micro-F1	AUC-PR
MusicCaps [†]	3,993	50	0.914	0.747	0.695	0.736	0.731
GTZAN [‡]	800	10	0.536	0.554	0.480	0.451	0.523
MagnaTagATune	15,431	50	0.563	0.350	0.304	0.392	0.287
FMA-small	6,400	50	0.914	0.430	0.267	0.325	0.326
B1 random (range)	—	—	—	—	0.039–0.148	—	—

[†] proxy task: aspect phrases are quoted in the caption. [‡] text is generated from audio descriptors, not human-written.

Task 4 (contrastive). A dual encoder projects $\mathbf{g}$ and $\mathbf{t}$ into a shared 256-d space, ℓ_2 -normalised so the inner product is cosine similarity, and is trained with a symmetric InfoNCE objective [van den Oord et al., 2018, Radford et al., 2021]:

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(\text{sim}(\mathbf{g}_i, \mathbf{t}_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(\mathbf{g}_i, \mathbf{t}_j)/\tau)}. \quad (5)$$

The brief writes the loss in one direction; we average both directions, since the brief also asks for retrieval metrics in both directions and training one direction alone would handicap the other. τ is initialised to 0.07 and learned (CLIP-style); a flag restores a fixed τ matching the literal formula.

Baselines. All four required baselines are implemented: B1 random and majority-class tag predictors, B2 a CNN over the mel-spectrogram (four conv blocks, global average pooling), B3 BERT-only, and B4 PCA(64) + MLP on mean-pooled hand-crafted features.

5 Experimental Setup

All experiments run on a single NVIDIA RTX 4070 Laptop GPU with PyTorch 2.10 (CUDA 12.6), PyTorch Geometric 2.8 [Fey and Lenssen, 2019], transformers 5.17 and librosa 0.10 [McFee et al., 2015]. Every run is seeded (42), trains for at most 20 epochs, and stops early once the validation metric has not improved for 4 epochs (with a per-task floor before early stopping may fire). **The best-validation checkpoint is always restored before test evaluation**, and per-tag decision thresholds are tuned on validation only. Test data is never used for model selection or threshold tuning. The full pipeline—preprocessing, all four tasks and all cross-dataset comparisons—takes ≈ 3.4 h end to end.

Metrics follow the brief: macro-F1 (mean over tags), micro-F1 (pooled), AUC-PR (mean average precision over tags), MAE/ R^2 for valence and arousal, and R@ K with median rank for retrieval.

6 Results

6.1 Task 1: text alone

Table 2 shows a 0.43 spread in test macro-F1 across datasets for an identical model. Almost all of it is explained by what the text is. MusicCaps captions contain the target phrases, so the model need only extract them. GTZAN’s generated descriptors are weakly but genuinely predictive of genre (0.480 vs. 0.148 random). MagnaTagATune and FMA-small supply only artist and album strings, and 0.30/0.27 is close to the ceiling of what identity metadata can say about 50 tags. FMA-small shows the largest generalisation gap (0.914 train vs. 0.267 test): with artist-disjoint official splits, memorised artist names transfer to nothing.

6.2 Task 2: audio structure alone

On GTZAN, GAT reaches 0.790 macro-F1 and beats the CNN baseline by 19.3 points; all three graph convolutions beat the CNN. On FMA-small the ordering inverts and the CNN wins narrowly (0.410 vs. 0.392 for the best GNN).

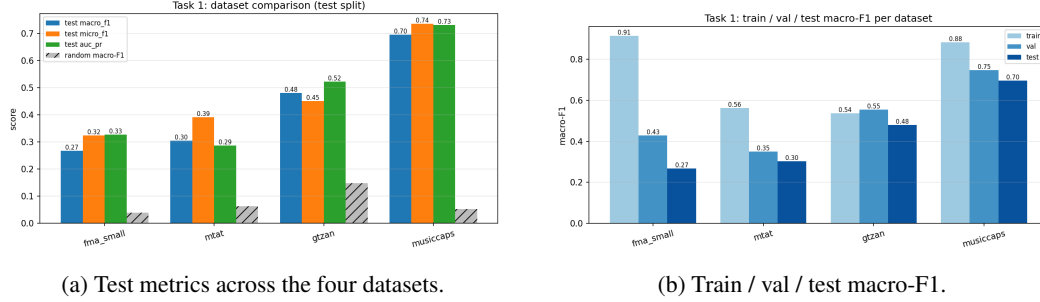


Figure 1: Task 1 across datasets. The right panel isolates the FMA-small generalisation gap: training macro-F1 of 0.91 collapses to 0.27 at test once the split is artist-disjoint.

Table 3: Task 2, genre classification from music-structure graphs, versus all required baselines. Best neural model per dataset in bold. The verdict reverses between datasets.

Model	GTZAN (10 genres, 799/100)				FMA-small (8 genres, 6397/800)			
	Acc	Macro-F1	Micro-F1	AUC-PR	Acc	Macro-F1	Micro-F1	AUC-PR
B1 random	0.110	0.112	0.110	0.147	0.151	0.151	0.151	0.135
B1 majority	0.100	0.018	0.100	0.100	0.125	0.028	0.125	0.125
B4 PCA+MLP	0.750	0.755	0.750	0.836	0.378	0.375	0.378	0.375
B2 CNN mel	0.610	0.597	0.610	0.735	0.406	0.410	0.406	0.446
GraphSAGE	0.730	0.723	0.730	0.827	0.400	0.392	0.400	0.434
GAT	0.800	0.790	0.800	0.883	0.395	0.388	0.395	0.422
GCN	0.750	0.753	0.750	0.861	0.389	0.380	0.389	0.412

The GTZAN confusion matrix (Figure 2a) localises the remaining error almost entirely in one class. Six of ten genres reach 0.90 recall and *metal* and *pop* reach 0.80, but *rock* collapses to 0.30 and is dispersed across five other genres—country, hip-hop, jazz, metal and pop (0.10–0.20 each). Rock is the least acoustically distinctive category in GTZAN and overlaps every neighbouring genre, so this is the expected failure mode rather than a defect of the graph representation; *reggae* (0.70) is the only other class below 0.80.

This comparison carries one caveat that we state explicitly because it cuts against our own result. The CNN is the slowest learner in the set: in an earlier run with a 60-epoch budget it was still improving at epoch 45 and reached 0.801 macro-F1 on GTZAN, versus 0.597 under the 20-epoch budget used here. The GNNs converge well inside 20 epochs. **The GTZAN margin in Table 3 is therefore inflated by the shortened schedule;** at equal convergence the honest GTZAN comparison is roughly GAT 0.838 versus CNN 0.801, a much narrower win. The FMA-small result, where the CNN wins despite being undertrained, is correspondingly a *stronger* result for the CNN than it appears.

6.3 Task 3: fusion and ablation

Table 4 is the centre of this report, and it does not say what the brief’s illustrative table anticipates. Three distinct orderings appear:

- **DEAM:** cross-attention wins clearly (0.429), and fusion beats both single modalities by a wide margin (+0.147 over BERT-only, +0.248 over GNN-only). This is the textbook result.
- **MagnaTagATune:** *gnn_only* is the *best* model (0.390), ahead of cross-attention (0.352) and far ahead of BERT-only (0.310). Adding text actively hurts. MagnaTagATune’s tags are instrument and texture descriptors—*guitar*, *strings*, *quiet*, *fast*, *no vocals*—which the audio graph captures directly, while its “text” is only artist/album identity that pulls the model toward memorisation.
- **FMA-small:** early concat wins (0.318) and cross-attention is the *worst* of the three fusion variants (0.205), below even BERT-only. With 6,400 training tracks and a 50-tag space, the extra attention parameters appear to overfit: the corresponding training macro-F1 is 0.883 against 0.205 at test.

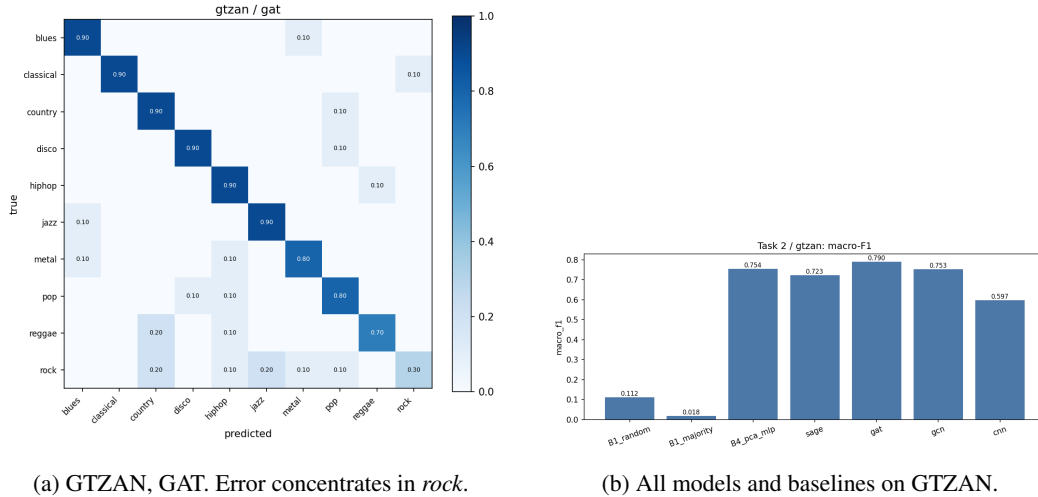


Figure 2: Task 2 on GTZAN. Every graph convolution beats the CNN mel baseline under this schedule, but see the convergence caveat in the text.

Table 4: Task 3 ablation, test macro-F1 / AUC-PR. Best per dataset in bold. The ranking of fusion strategies is *not* stable across datasets.

Fusion	FMA-small (50 tags)		MagnaTagATune (50 tags)		DEAM (20 tags)	
	Macro-F1	AUC-PR	Macro-F1	AUC-PR	Macro-F1	AUC-PR
B1 random	0.039	0.045	0.064	0.065	0.085	0.098
gnn_only	0.183	0.181	0.390	0.368	0.181	0.194
bert_only	0.258	0.330	0.310	0.296	0.282	0.380
concat	0.318	0.369	0.349	0.339	0.299	0.402
cross_attn	0.205	0.299	0.352	0.346	0.429	0.452

The auxiliary emotion head on DEAM gives the cleanest evidence that the graph carries information the text does not (Table 5). Valence R^2 is 0.52 for cross-attention and 0.40 for GNN-only, but -0.13 for BERT-only—that is, a text-only model is *worse than predicting the mean*. Track titles and artist names simply do not encode arousal; the audio does.

Figure 3 projects the fused representation $\mathbf{z}$ for the DEAM test split. The structure is coarse but real: the left region is dominated by acoustic and traditional genres (classical, blues, folk, jazz, country) and the right region by produced genres (rock, electronic, hip-hop, pop, soul/R&B). Individual genres are *not* linearly separated, which is what a test macro-F1 of 0.43 over 20 tags should look like. Colouring the same projection by mood quadrant shows partial arousal structure along the same left–right axis and no clean valence separation—so the dominant axis of $\mathbf{z}$ appears to be energy/production rather than genre identity.

6.4 Task 4: contrastive cross-modal retrieval

On MusicCaps the dual encoder retrieves the correct clip in the top 5 for 19.3% of captions against a 0.97% random reference—roughly $20\times$ chance, with a median rank of 25 out of 513. On DEAM the same architecture reaches only $1.8\times$ chance. The difference is the text: MusicCaps pairs each clip with an expert description of *what the audio sounds like*, whereas DEAM’s text is title and artist, which is close to unpredictable from audio. This is the same signal-source explanation that governs Tasks 1–3.

The validation recall curve (Figure 4a) shows $R@5$ and $R@10$ rising steeply for the first ≈ 8 epochs and then plateauing, while $R@1$ stays flat at 0.05–0.08 throughout. The model learns to place the correct clip in a shortlist quickly but never learns to rank it first, which is the expected signature of a caption describing a *class* of sounds that many clips in the pool satisfy equally well.

Table 5: Task 3 auxiliary emotion regression on DEAM (targets rescaled to $[-1, 1]$). The text-only model fails outright.

Fusion	MAE valence	MAE arousal	R^2 valence	R^2 arousal
bert_only	0.254	0.271	-0.129	-0.089
gnn_only	0.192	0.200	0.397	0.391
concat	0.186	0.193	0.435	0.436
cross_attn	0.167	0.201	0.521	0.334

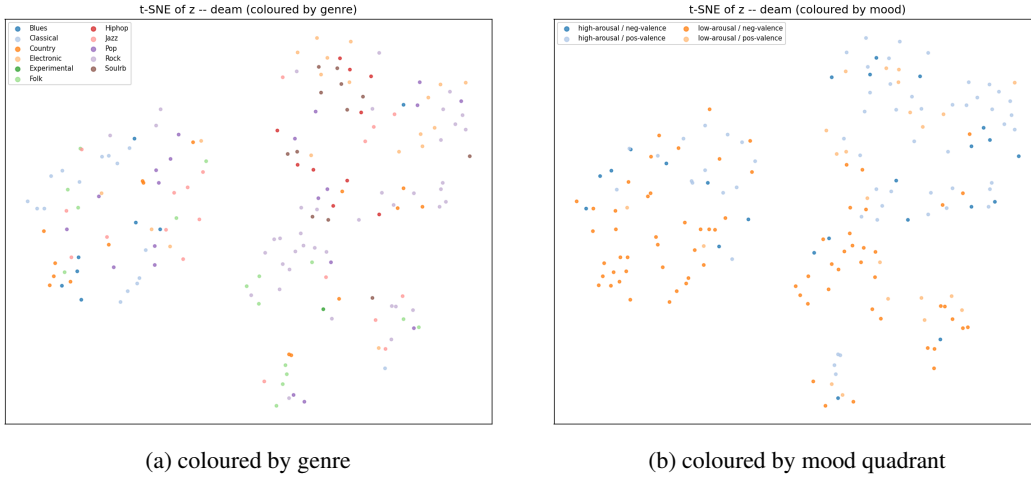


Figure 3: t-SNE [van der Maaten and Hinton, 2008] of the fused representation $\mathbf{z}$ on the DEAM test split, cross-attention model. Both panels split along the same left-right axis, suggesting $\mathbf{z}$ is organised primarily by energy/production rather than by genre label.

Zero-shot tag prediction from the contrastive space—embedding the prompt “a music clip that sounds {tag}” and ranking tags by similarity—reaches 0.129 macro-F1 on DEAM against 0.429 for the Task 3 supervised model, a 0.30 gap; on MusicCaps it reaches 0.124. Contrastive alignment learned from 1,431–4,134 pairs is not a substitute for supervision at this scale.

6.5 Cross-dataset synthesis

Pulling the four tasks together, one rule organises almost every result: **music-structure graphs help exactly to the extent that the label space describes musical structure, and text helps exactly to the extent that it describes sound rather than identity**. MagnaTagATune’s instrument and texture tags are structural, so GNN-only wins there. DEAM’s emotion targets are acoustic, so the graph is indispensable and text alone fails. MusicCaps captions describe sound, so both the BERT tagger and the contrastive encoder do well. FMA and GTZAN genre labels depend heavily on production timbre, which the mel-CNN reads more directly than a 19-node segment graph does.

7 Discussion and Limitations

The graph-coherence diagnostic saturates. The brief proposes $S_{\text{graph}} = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} \mathbb{1}[\cos(\mathbf{h}_i, \mathbf{h}_j) > \tau]$ as an analysis of whether high-attention edges align with repeated patterns. We implemented and report it, but it returns 0.996–1.000 on every dataset and every fusion mode, so it discriminates nothing. The cause is that $\mathbf{h}$ is taken after batch-norm + ReLU, which makes activations non-negative and drives pairwise cosine similarity high regardless of structure. A useful version would need a higher τ , pre-activation features, or a comparison against a degree-preserving rewired null graph. We report this as a negative result rather than present a saturated statistic as evidence.

Table 6: Task 4 retrieval on the held-out test split. The random reference is K/N for a pool of N test items.

Dataset	Direction	R@1	R@5	R@10	Median rank	Pool N
MusicCaps	caption $\rightarrow$ audio	0.066	0.193	0.300	25	513
	audio $\rightarrow$ caption	0.049	0.187	0.298	25	513
	random reference	0.002	0.010	0.020	—	513
DEAM	caption $\rightarrow$ audio	0.000	0.051	0.107	68	177
	audio $\rightarrow$ caption	0.011	0.057	0.096	62	177
	random reference	0.006	0.028	0.057	—	177

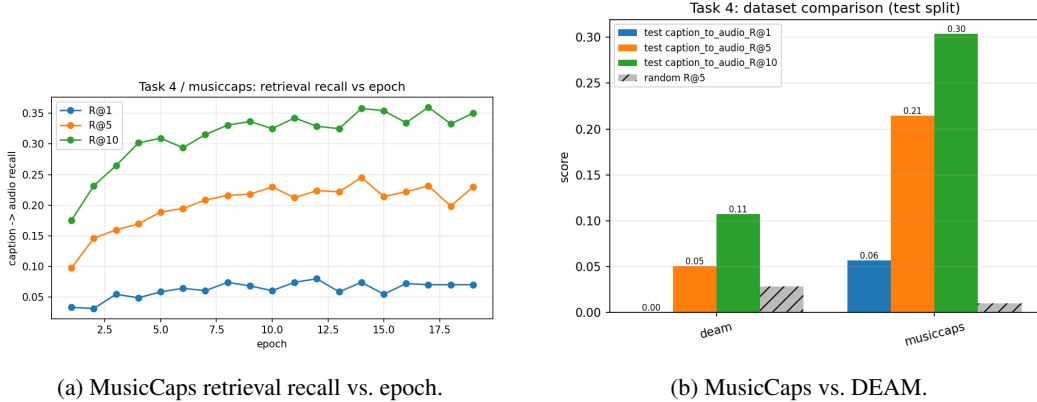


Figure 4: Task 4. MusicCaps reaches $\approx 20\times$ the random reference; DEAM, whose text is title and artist rather than a description of the sound, reaches only $\approx 1.8\times$.

Schedule length biases the baseline comparison. As noted in Section 6.2, the 20-epoch budget adopted for tractability understates the CNN baseline by roughly 0.20 macro-F1 on GTZAN. Any claim that graphs beat CNNs should be read against the 60-epoch figures quoted there, not the headline table.

Run-to-run variance is non-trivial on small test sets. DEAM was trained twice under identical settings (once inside the comparison sweep, once to save a checkpoint) and produced 0.414 and 0.429 test macro-F1. With 177 test clips, differences below ~ 0.02 should not be interpreted. GTZAN, with 100 test clips, is worse. Single-seed results are a genuine limitation of this study; a proper treatment would report mean $\pm$ std over at least three seeds.

Case studies show the text channel is weak where text is identity. Inspecting cross-attention on DEAM, the tokens the graph attends to on a correctly-structured track are frequently uninformative fragments of the artist name (e.g. & at weight 0.64, smith at 0.26). The chord paths themselves are musically sensible—one test track yields $A:maj \rightarrow D:maj \rightarrow A:maj$, a I–IV–I motion in A—so the failure is in the text channel, not the graph. All three DEAM case studies predicted a plausible but wrong genre for *Blues* tracks, which is consistent with metadata carrying little genre signal.

Other limitations. We use only the first 30 s of each track and cap graphs at 24 nodes, so long-range form (verse/chorus repetition across minutes) is invisible. Chord estimation is template matching on chroma, far below the accuracy of a trained chord recogniser. The MagnaTagATune official split is not artist-disjoint. The MusicCaps Task 1 result is a proxy by construction, and the GTZAN Task 1 text is machine-generated. The human listening evaluation required by the brief is prepared—a sheet of 10 retrieval pairs with five rating columns is generated automatically—but has not yet been filled in by listeners, so no human scores are reported here.

8 Conclusion

We built and evaluated a full GNN–BERT pipeline for music context understanding across five datasets and four tasks. Every model beats its required baselines by large margins, and the auxiliary

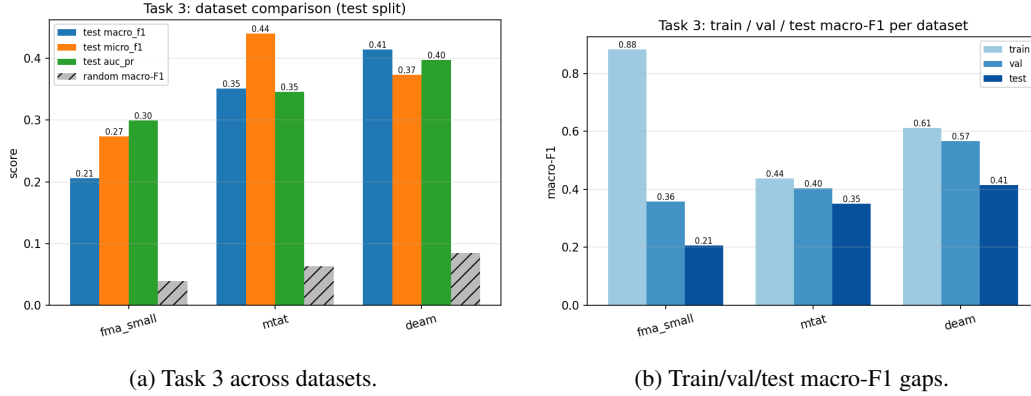


Figure 5: Cross-dataset comparison for Task 3. The right panel shows that FMA-small’s poor test score is an overfitting problem, not an underfitting one.

emotion result ($R^2 = 0.52$ with the graph versus -0.13 without) is direct evidence that audio structure carries context that text metadata cannot. But the headline lesson from the cross-dataset design is a negative one for any single architecture: cross-attention fusion is best on one of three datasets, early concatenation on another, and a text-free GNN on the third. Reporting only the dataset where cross-attention wins would have produced a cleaner but misleading paper. The practical recommendation is to choose the fusion strategy from the semantics of the label space—structural labels favour the graph, descriptive text favours fusion, identity-only metadata favours dropping text altogether.

Acknowledgments and Disclosure of Funding

We thank the course instructor for the project specification. All experiments were run on a single consumer laptop GPU; the complete pipeline, configuration and per-experiment metric files are included with this submission.

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A Reproducibility

The complete pipeline is reproduced by two commands:

```
python -m src.prepare_data --stage all --jobs 10 # text, splits, 56,589 graphs
.\scripts\run_all_tasks.ps1 # tasks 1-4 + comparisons
```

Artefacts: `results/metrics/` holds one JSON per experiment containing the full argument namespace, per-epoch history, restored best epoch, tuned thresholds, test metrics and baselines; `results/comparison/` holds the cross-dataset tables and charts; `results/plots/` holds F1 curves, confusion matrices and t-SNE projections; `results/case_studies/` holds Task 3 chord paths and text alignments; `results/retrieval_examples/` holds the Task 4 qualitative examples and the listening-test sheet. At least 20 preprocessed graphs per dataset are exported as both `.pt` tensors and readable `.json` under `data/processed/graph_samples/`.

B Wall-clock cost

Preprocessing: GTZAN 1.1 min, DEAM 2.0 min, MusicCaps 4.0 min, FMA-small $\approx$ 10 min, MagnaTagATune $\approx$ 25 min (10 worker processes). Training (Table 7) totals 202 min.

Table 7: Wall-clock time per pipeline step on one RTX 4070 Laptop GPU.

Step	Minutes
Task 1 comparison (4 datasets)	41.0
Task 1 attention visualisation	7.1
Task 2 comparison (2 datasets $\times$ 6 models)	5.6
Task 3 comparison (3 datasets $\times$ 4 fusions)	129.5
Task 3 checkpoint run	6.9
Task 4 comparison (2 datasets)	6.7
Task 4 checkpoint run	5.3
Total	202.2